

Geometry-based Methods in Vision

CMU 16-822, Fall 2024

Instructor: Shubham Tulsiani

Administrivia

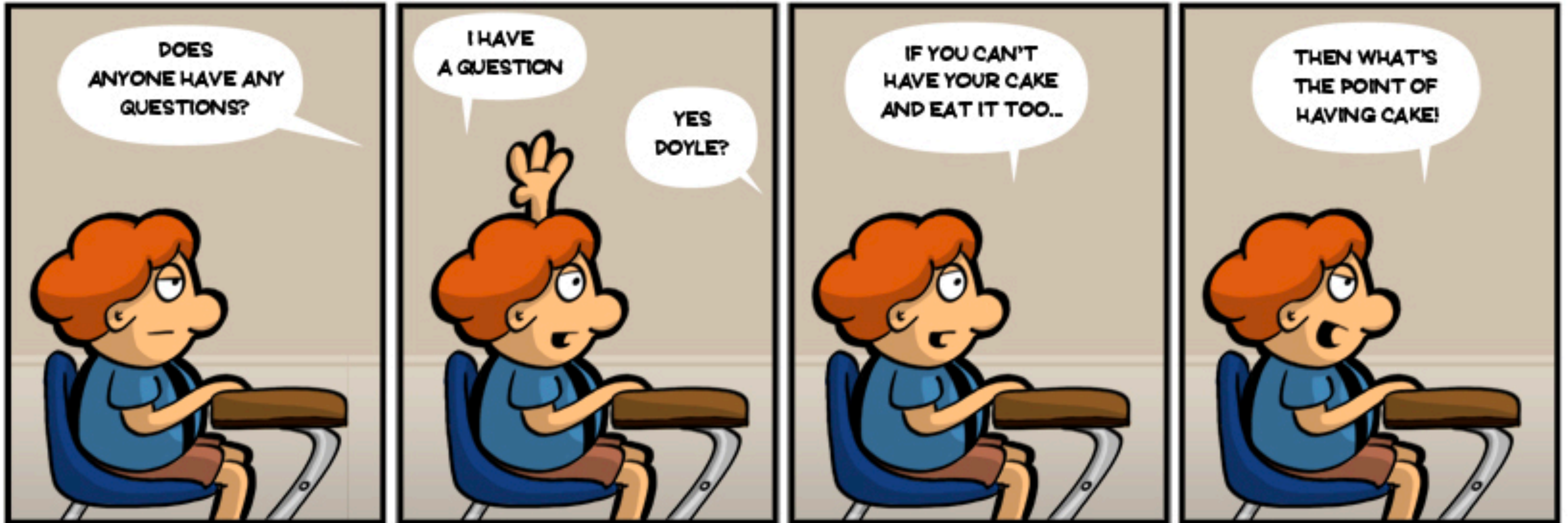
- Assignment 2
 - *Due today*
- Project Proposals
 - *Please form teams (on canvas)*
 - *Due Saturday (**no late days**)*
- PS3 and Assignment 3
 - *Released today*
 - *Due in 2 and 3 weeks respectively*
- No office hours in Fall Break

Lecture 13: Auto-calibration

A reminder to ask questions!

UNDERWHELMED

1/16/09



"Any questions?"

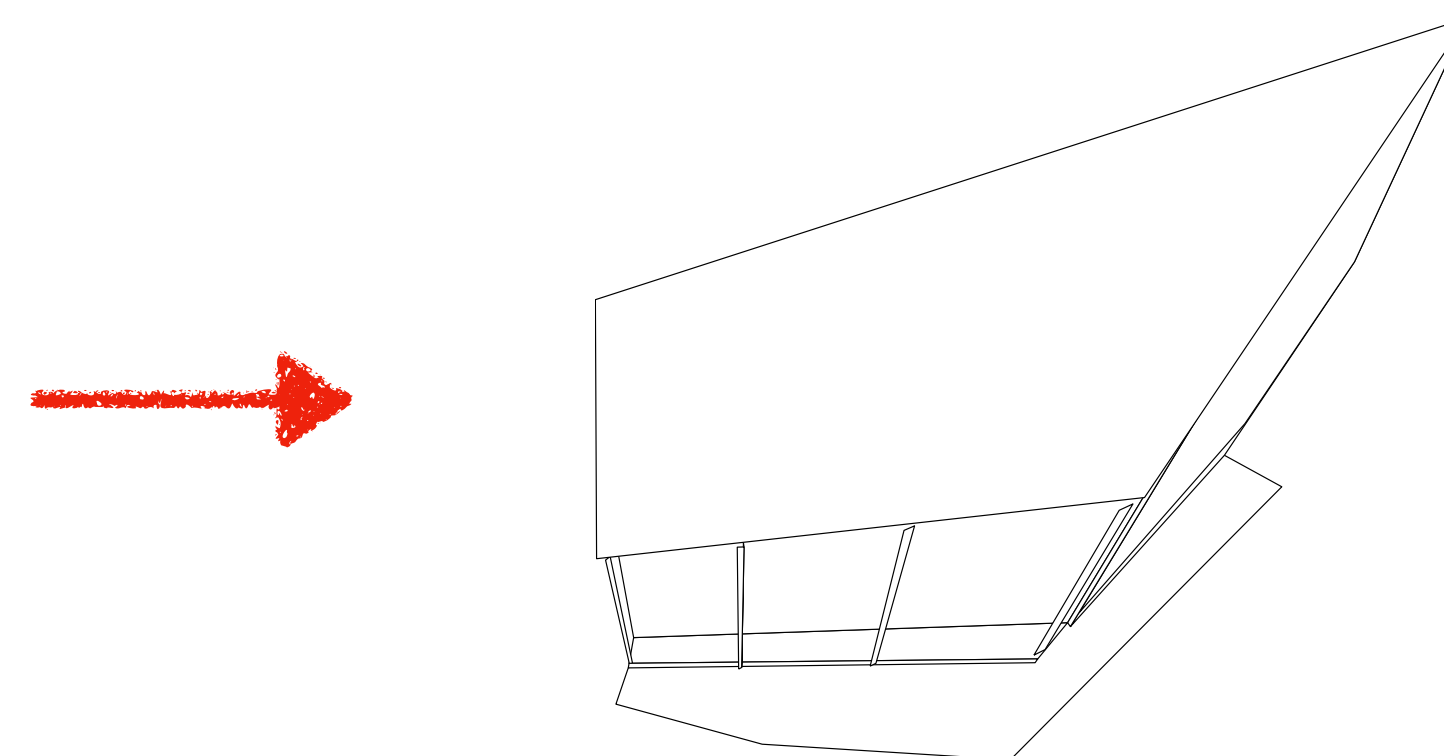
© 2009 Sean McLean Underwhelmedcomic.com

Projective-ambiguities in 3D from Two views

$$\begin{array}{ccc} \{(\mathbf{x}_i, \mathbf{x}'_i)\} & \longrightarrow & (\mathbf{P}, \mathbf{P}') \quad \{\mathbf{X}_i\} \\ \{(\mathbf{x}_i, \mathbf{x}'_i)\} & \longrightarrow & (\mathbf{P}H, \mathbf{P}'H) \quad \{H^{-1}\mathbf{X}_i\} \end{array}$$

Calibrated case: can only recover 3D up to a similarity transform as $(\mathbf{P}, \mathbf{P}')$ are ambiguous up to a similarity

Uncalibrated case: can only recover 3D up to a homography as $(\mathbf{P}, \mathbf{P}')$ are ambiguous up to a general H



Projective-ambiguities in 3D from Multi-view

$$\{(\mathbf{x}_{n_1}, \mathbf{x}_{n_2}, \dots, \mathbf{x}_{n_k})_i\} \longrightarrow \{\mathbf{P}_n\} \quad \{\mathbf{X}_i\}$$

corresponding observation
across subset of images

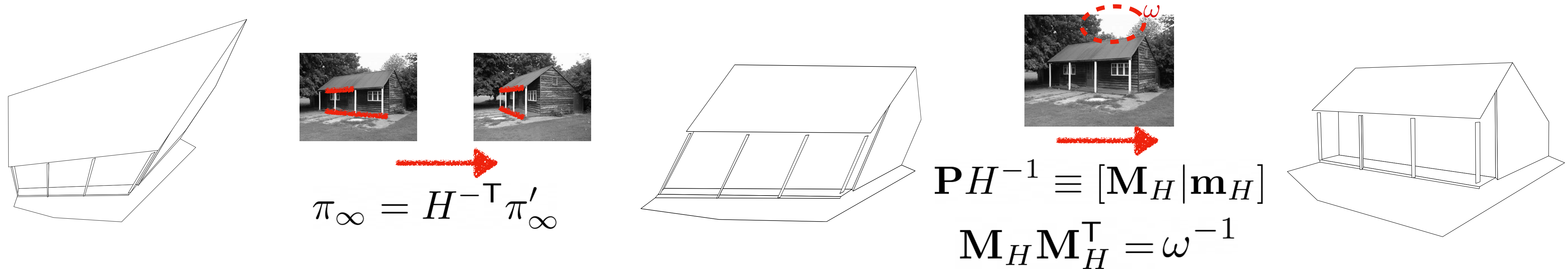
$$\{(\mathbf{x}_{n_1}, \mathbf{x}_{n_2}, \dots, \mathbf{x}_{n_k})_i\} \longrightarrow \{\mathbf{P}_n H\} \quad \{H^{-1} \mathbf{X}_i\}$$

3D reconstruction from multi-view from 2D correspondences — ambiguities up to a homography H
(similar to two-view case)

$$\mathbf{x}_{i,n} = \mathbf{P}_n \mathbf{X}_i = (\mathbf{P}_n H)(H^{-1} \mathbf{X}_i)$$

Transformations of cameras and (inverse) transformation for 3D points yields **precisely** same 2D projections

Review — Rectification from Image Annotations



Possible to rectify given: a) the plane at infinity in 3D, and b) the IAC in one image
(both obtained via user annotations)

Auto-calibration : Can we recover metric reconstruction **without** user annotations?
(using only simple constraints on the intrinsic matrices)

Auto-calibration: Setup

$$\{\mathbf{P}_n\} \quad \{\mathbf{X}_i\}$$
$$\{(\mathbf{x}_{n_1}, \mathbf{x}_{n_2}, \dots, \mathbf{x}_{n_k})_i\}$$

Input: 3D Reconstruction and
corresponding cameras



$$H$$

Output: Homography that metrically
rectifies the reconstruction

How is this even possible (given no annotations about parallelism/orthogonality)?

Auto-calibration: Intuition

Let's examine the camera intrinsics before/after a transformation by a homography H
 (for some random H , and cameras defined with random R, t)

K1
 $\begin{bmatrix} 10 & 0 & 5 \\ 0 & 8 & 4 \\ 0 & 0 & 1 \end{bmatrix}$

K2
 $\begin{bmatrix} 10 & 0 & 6 \\ 0 & 9 & 4 \\ 0 & 0 & 1 \end{bmatrix}$

K3
 $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\xrightarrow{H}$

$$\mathbf{P}'_i = \mathbf{P}_i H$$

$\xrightarrow[\text{decomp.}]{\text{RQ}}$

$$K'_i$$

K1
 $\begin{bmatrix} 5.69576642 & 15.2432634 & -4.24290847 \\ 0. & 4.62393363 & 5.59888405 \\ 0. & 0. & 0.64776583 \end{bmatrix}$

K2
 $\begin{bmatrix} 6.89730821 & -16.52582543 & 2.16534377 \\ 0. & -15.90040989 & -2.2728933 \\ 0. & 0. & -0.37531846 \end{bmatrix}$

K3
 $\begin{bmatrix} 0.3161438 & 0.48533096 & 0.08237905 \\ 0. & 1.00713676 & 0.38479716 \\ 0. & 0. & 0.66342748 \end{bmatrix}$

$$\mathbf{P}_i = K_i [R_i | t_i]$$

structural properties (e.g. equal focal length, zero skew) are not preserved!

Auto-calibration: Intuition

K1
[[5.69576642 15.2432634 -4.24290847]
[0. 4.62393363 5.59888405]
[0. 0. 0.64776583]]

K2
[[6.89730821 -16.52582543 2.16534377]
[0. -15.90040989 -2.2728933]
[0. 0. -0.37531846]]

K3
[[0.3161438 0.48533096 0.08237905]
[0. 1.00713676 0.38479716]
[0. 0. 0.66342748]]



$\{K_i\}$

General approach to auto-calibration — find H s.t. some known properties of intrinsics (e.g. zero skew, or square pixels) are reflected

Auto-calibration: Setup

$$\{\mathbf{P}_n\} \quad \{\mathbf{X}_i\}$$
$$\{(\mathbf{x}_{n_1}, \mathbf{x}_{n_2}, \dots, \mathbf{x}_{n_k})_i\}$$

Input: Recovered Camera Matrices



$$\mathbf{H}$$

Output: Homography that metrically
rectifies the reconstruction

We'll not leverage any information about the metric structure (only aspects of camera intrinsics)

Auto-calibration: Setup

$$\{\mathbf{P}_n\} \xrightarrow[H]{\text{red arrow}} \{\mathbf{P}'_n \equiv \mathbf{P}_n H\}$$

We can assume a 'canonical' form for the solution:

$$\mathbf{P}_1 = [I|\mathbf{0}]$$

$$\mathbf{P}'_1 = K_1 [I|\mathbf{0}]$$

~~we are not assuming known K_1 , just~~
centering the coordinate system

If $\mathbf{P}'_1$ is not in canonical form, how can we modify the solution H to enforce this?

$$\mathbf{P}'_1 = K_1 [R|\mathbf{t}] \quad H_c = H \begin{bmatrix} R^\top & -R^\top \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \quad \begin{array}{l} \{\mathbf{P}'_n \equiv \mathbf{P}_n H\} \\ \{\mathbf{P}'_n \equiv \mathbf{P}_n H_c\} \end{array} \quad \begin{array}{l} \text{Same intrinsics in} \\ \text{both cases} \end{array}$$

Auto-calibration: Setup

$$\begin{array}{ccc} \{\mathbf{P}_n\} & \xrightarrow{H} & \{\mathbf{P}'_n \equiv \mathbf{P}_n H\} \\ \mathbf{P}_1 = [I|\mathbf{0}] & & \mathbf{P}'_1 = K_1 [I|\mathbf{0}] \end{array}$$

General case:

$$H = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^\top & k \end{bmatrix}$$

Canonical form:

$$H = \begin{bmatrix} K_1 & \mathbf{0} \\ \mathbf{v}^\top & 1 \end{bmatrix} \quad \mathbf{t} = \mathbf{0}.$$

Can also set $k = 1$

Q: How many degrees of freedom to estimate?

5 + 3

Auto-calibration: Interpreting $\mathbf{v}$

$$\begin{array}{ccc} \{\mathbf{P}_n\} & H = \begin{bmatrix} K_1 & \mathbf{0} \\ \mathbf{v}^\top & 1 \end{bmatrix} & \{\mathbf{P}'_n \equiv \mathbf{P}_n H\} \\ \mathbf{P}_1 = [I | \mathbf{0}] & \xrightarrow{\text{red arrow}} & \mathbf{P}'_1 = K_1 [I | \mathbf{0}] \end{array}$$

$$\mathbf{v} = -K_1^\top \mathbf{p}$$

$$\pi_\infty = (\mathbf{p}^\top, 1)^\top$$

plane at infinity in current
projective reconstruction

Suppose H metrically rectifies the reconstruction:

$$\mathbf{X}'_i = H^{-1} \mathbf{X}_i$$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = H^\top \pi_\infty$$

$$\pi_\infty = H^{-\top} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{bmatrix} (K^1)^{-\top} & -(K^1)^{-\top} \mathbf{v} \\ \mathbf{0} & 1 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -(K^1)^{-\top} \mathbf{v} \\ 1 \end{pmatrix}$$

Auto-calibration: Interpreting $\mathbf{v}$

$$\begin{array}{ccc} \{\mathbf{P}_n\} & H = \begin{bmatrix} K_1 & \mathbf{0} \\ \mathbf{v}^\top & 1 \end{bmatrix} & \{\mathbf{P}'_n \equiv \mathbf{P}_n H\} \\ \mathbf{P}_1 = [I|\mathbf{0}] & \xrightarrow{\text{red arrow}} & \mathbf{P}'_1 = K_1 [I|\mathbf{0}] \end{array}$$

$$\mathbf{v} = -K_1^\top \mathbf{p}$$

$$\pi_\infty = (\mathbf{p}^\top, 1)^\top$$

plane at infinity in current
projective reconstruction

Can compute H given: a) plane at infinity, and b) camera calibration for one camera

Q: where did we see this before?

Auto-calibration: Basic Equations

$$\begin{array}{ccc}
 \{\mathbf{P}_n\} & H = \begin{bmatrix} K_1 & \mathbf{0} \\ \mathbf{v}^\top & 1 \end{bmatrix} & \{\mathbf{P}'_n \equiv \mathbf{P}_n H\} \\
 \mathbf{P}_1 = [I | \mathbf{0}] & \xrightarrow{\mathbf{A} \equiv \begin{bmatrix} K_1 \\ \mathbf{v}^\top \end{bmatrix}} & \mathbf{P}'_1 = K_1 [I | \mathbf{0}] \\
 & & \mathbf{P}'_n = K_n R_n [I | \tilde{\mathbf{C}}_n]
 \end{array}$$

$$K_n R_n = \mathbf{P}_n \mathbf{A}$$

$$K_n K_n^\top = \mathbf{P}_n \underbrace{\mathbf{A} \mathbf{A}^\top}_{\mathbf{L}} \mathbf{P}_n^\top$$

Reminder:

$$\omega^{-1} = K K^\top \equiv \omega^*$$

This matrix is related to the dual of image of Absolute Conic,
(but for now, we can just assume this as a notation convenience)

Auto-calibration: Basic Equations

$$\begin{array}{ccc} \{\mathbf{P}_n\} & H = \begin{bmatrix} K_1 & 0 \\ \mathbf{v}^\top & 1 \end{bmatrix} & \{\mathbf{P}'_n \equiv \mathbf{P}_n H\} \\ \mathbf{P}_1 = [I|\mathbf{0}] & \xrightarrow{\mathbf{A} \equiv \begin{bmatrix} K_1 \\ \mathbf{v}^\top \end{bmatrix}} & \mathbf{P}'_1 = K_1 [I|\mathbf{0}] \end{array}$$

$$\omega_n^* = \mathbf{P}_n \mathbf{L} \mathbf{P}_n^\top$$

Approach Overview:

1. Identify how information on K_n constrains ω_n^*
2. Obtain $\mathbf{L}$ via constraints on ω_n^*
3. Compute $\mathbf{A}$ from $\mathbf{L}$

Q: How many parameters to estimate in $\mathbf{L}$?
9 (10 in a symmetric matrix, but scale does not matter)

Q: Why is there a scale ambiguity again despite setting $k=1$ in H ?

Q: Why more parameters than $\mathbf{A}$?

Auto-calibration

Step 1: Identify how information on K_n constrains ω_n^*

$$\omega_n^* = \mathbf{P}_n \mathbf{L} \mathbf{P}_n^T$$

$$\mathbf{K} = \begin{bmatrix} \alpha_x & s & x_o \\ 0 & \alpha_y & y_o \\ 0 & 0 & 1 \end{bmatrix} \quad \omega^* = \mathbf{K} \mathbf{K}^T = \begin{bmatrix} \alpha_x^2 + s^2 + x_o^2 & s\alpha_y + x_o y_o & x_o \\ s\alpha_y + x_o y_o & \alpha_y^2 + y_o^2 & y_o \\ x_o & y_o & 1 \end{bmatrix}$$

Constraints with principal point

principal point at (0, 0): $\omega_{1,3}^* = \omega_{2,3}^* = 0$

principal point at (u, v): ~~$\omega_{1,3}^* = u; \omega_{2,3}^* = v$~~ $\omega_{1,3}^* = \omega_{3,3}^* u; \omega_{2,3}^* = \omega_{3,3}^* v$

+ zero skew (with known p.p): $\omega_{1,2}^* = \omega_{3,3}^* uv;$

+ equal focal lengths in x,y: $\omega_{1,1}^* - \omega_{2,2}^* = (u^2 - v^2)\omega_{3,3}^*;$

All of the above are linear constraints in ω_n^*

Auto-calibration

Step 1: Identify how information on K_n constrains ω_n^*

$$\omega_n^* = \mathbf{P}_n \mathbf{L} \mathbf{P}_n^\top$$

$$\mathbf{K} = \begin{bmatrix} \alpha_x & s & x_o \\ 0 & \alpha_y & y_o \\ 0 & 0 & 1 \end{bmatrix} \quad \omega^* = \mathbf{K} \mathbf{K}^T = \begin{bmatrix} \alpha_x^2 + s^2 + x_o^2 & s\alpha_y + x_o y_o & x_o \\ s\alpha_y + x_o y_o & \alpha_y^2 + y_o^2 & y_o \\ x_o & y_o & 1 \end{bmatrix}$$

General Constraints

$$\text{zero skew : } \omega_{1,2}^* \omega_{3,3}^* = \omega_{1,3}^* \omega_{2,3}^*$$

$$+ \text{ equal focal lengths in x,y : } (\omega_{1,1}^* - \omega_{2,2}^*) \omega_{3,3}^* = \omega_{1,3}^{*2} - \omega_{2,3}^{*2}$$

$$\text{shared intrinsics across cameras (i,j) : } \omega_{11}^{*i}/\omega_{11}^{*j} = \omega_{12}^{*i}/\omega_{12}^{*j} = \omega_{13}^{*i}/\omega_{13}^{*j} = \omega_{22}^{*i}/\omega_{22}^{*j} = \omega_{23}^{*i}/\omega_{23}^{*j} = \omega_{33}^{*i}/\omega_{33}^{*j}$$

Non-linear constraints in ω_n^* and $\mathbf{L}$

Auto-calibration

Step 2: Obtain $\mathbf{L}$ via constraints on ω_n^*

$$\omega_n^* = \mathbf{P}_n \mathbf{L} \mathbf{P}_n^T$$

principal point at (u, v): $\omega_{1,3}^* = \omega_{3,3}^* u; \quad \omega_{2,3}^* = \omega_{3,3}^* v$

zero skew :

$$\omega_{1,2}^* \omega_{3,3}^* = \omega_{1,3}^* \omega_{2,3}^*$$

+ zero skew (with known p.p): $\omega_{1,2}^* = \omega_{3,3}^* uv;$

+ equal f_x, f_y : $(\omega_{1,1}^* - \omega_{2,2}^*) \omega_{3,3}^* = \omega_{1,3}^{*2} - \omega_{2,3}^{*2}$

+ equal focal lengths in x,y: $\omega_{1,1}^* - \omega_{2,2}^* = (u^2 - v^2) \omega_{3,3}^*;$

shared intrinsics:

$$\begin{aligned} \omega_{11}^{*i} / \omega_{11}^{*j} &= \omega_{12}^{*i} / \omega_{12}^{*j} = \omega_{13}^{*i} / \omega_{13}^{*j} \\ &= \omega_{22}^{*i} / \omega_{22}^{*j} = \omega_{23}^{*i} / \omega_{23}^{*j} = \omega_{33}^{*i} / \omega_{33}^{*j} \end{aligned}$$

Linear constraints in ω_n^*  Linear constraints in $\mathbf{L}$ i.e. $\mathbf{C}\mathbf{l} = 0; \quad \|\mathbf{l}\| = 1$

Q: How many views required in each case for determining $\mathbf{L}$?

Need 9 constraints to solve linear system (without using $\det(\mathbf{L})=0$)

Can solve with 8 constraints if enforcing $\det(\mathbf{L})=0$

Q: where did we do something like this before?

Auto-calibration

Step 2: Obtain $\mathbf{L}$ via constraints on $\boldsymbol{\omega}_n^*$

$$\boldsymbol{\omega}_n^* = \mathbf{P}_n \mathbf{L} \mathbf{P}_n^T$$

principal point at (u, v):

$$\omega_{1,3}^* = \omega_{3,3}^* u; \quad \omega_{2,3}^* = \omega_{3,3}^* v$$

zero skew :

$$\omega_{1,2}^* \omega_{3,3}^* = \omega_{1,3}^* \omega_{2,3}^*$$

+ zero skew (with known p.p):

$$\omega_{1,2}^* = \omega_{3,3}^* uv;$$

+ equal f_x, f_y : $(\omega_{1,1}^* - \omega_{2,2}^*) \omega_{3,3}^* = \omega_{1,3}^{*2} - \omega_{2,3}^{*2}$

+ equal focal lengths in x,y :

$$\omega_{1,1}^* - \omega_{2,2}^* = (u^2 - v^2) \omega_{3,3}^*;$$

shared intrinsics:

$$\begin{aligned} \omega_{11}^{*i} / \omega_{11}^{*j} &= \omega_{12}^{*i} / \omega_{12}^{*j} = \omega_{13}^{*i} / \omega_{13}^{*j} \\ &= \omega_{22}^{*i} / \omega_{22}^{*j} = \omega_{23}^{*i} / \omega_{23}^{*j} = \omega_{33}^{*i} / \omega_{33}^{*j} \end{aligned}$$

Non- linear constraints in $\boldsymbol{\omega}_n^*$



Iterative non-linear optimization for $\mathbf{L}$

(or, directly optimize $\mathbf{A}$ as $\mathbf{L} = \mathbf{A} \mathbf{A}^T$)

Q: How many views required in each case for determining $\mathbf{L}/\mathbf{A}$?

Auto-calibration

Step 2: Obtain L via constraints on ω_n^*

Condition	fixed f	known k	views m
Constant internal parameters	5	0	3
Aspect ratio and skew known, focal length and principal point vary	0	2	4*
Aspect ratio and skew constant, focal length and principal point vary	2	0	5*
Skew zero, all other parameters vary	0	1	8*
p.p. known all other parameters vary	0	2	4*, 5(linear)
p.p. known skew zero	0	3	3(linear)
p.p., skew and aspect ratio known	0	4	2, 3(linear)

Table 19.3. *The number of views m required under various conditions in order for there to be enough constraints for auto-calibration. For those cases marked with an asterisk there may be multiple solutions, even for general motion between views.*

Auto-calibration

Step 3: Compute $\mathbf{A}$ from $\mathbf{L}$ ($\mathbf{L} = \mathbf{A}\mathbf{A}^\top$)

$$\mathbf{L} = UDU^\top \xrightarrow{\det(\mathbf{L})=0} \mathbf{L} = U\bar{D}U^\top \quad \bar{D} = \text{diag}(D_{11}, D_{22}, D_{33}, 0)$$

~~$$\mathbf{A} = U[:, 4, : 3] (\bar{D}[:, 3, : 3])^{0.5}$$~~

(Does not follow the structure of $\mathbf{K}$) $\mathbf{A} \equiv \begin{bmatrix} K_1 \\ \mathbf{v}^\top \end{bmatrix}$

$$\mathbf{L} = \begin{bmatrix} K_1 K_1^\top & K_1 \mathbf{v} \\ (K_1 \mathbf{v})^\top & \mathbf{v}^\top \mathbf{v} \end{bmatrix}$$

a) Compute K_1 via Cholesky decomposition of 3 X 3 sub-matrix of $\mathbf{L}$

b) Compute $\mathbf{v}$ given $\mathbf{L}$ and K_1

Auto-calibration: Summary

$$\begin{array}{ccc} \{\mathbf{P}_n\} & \{\mathbf{X}_i\} & \\ \{(\mathbf{x}_{n_1}, \mathbf{x}_{n_2}, \dots, \mathbf{x}_{n_k})_i\} & \xrightarrow{\text{red arrow}} & H = \begin{bmatrix} K_1 & \mathbf{0} \\ \mathbf{v}^\top & 1 \end{bmatrix} \end{array}$$

Input: Recovered Camera Matrices

Output: Homography that metrically
rectifies the reconstruction

Can recover metric reconstruction in several common cases **without** image annotations:

- 3 or more images from the same camera
- 8 (or more) images with zero skew, or 4 images with square pixels
- 2 images with known p.p, square pixels
- and others...

Questions?